Homework #3

Problems 6 and 7 are graphical solutions. Please use a protractor, ruler, and compass and work as accurately as possible. Alternatively you could use a vector drawing program to work out the geometry.

1) A stone released at the top of the shaft of a well hits the water 1.55 s later. (a) How deep is the well? (b) At what speed does the stone hit the water?

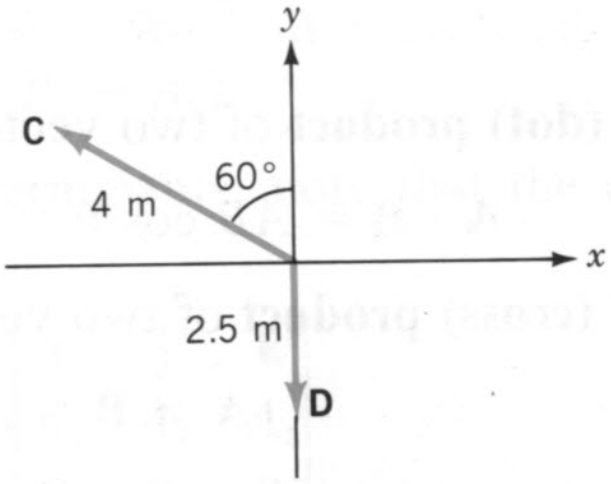
2) A baseball is thrown up at 30.0 m/s from the ground. Find: (a) its velocity at a height of 25.0 m; (b) when its speed is 15.0 m/s; (c) when its height is 40.0 m.

3) A ball thrown up from a rooftop of height 40.0 m lands on the ground in 4.10 s. (a) What is the balls maximum height above the ground? (b) What is its velocity 15.0 m below the rooftop?

4) A projectile is fired vertically up from the ground. Find its maximum height and the time of flight given that it loses 30% of its initial speed in 1.80 s and is moving upward.

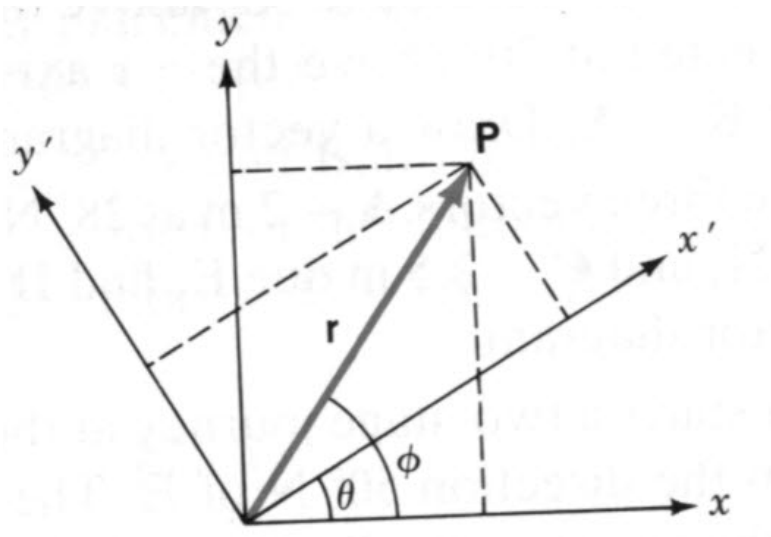
5) Ball A is thrown up from the ground at 25.0 m/s and ball B is thrown down at 15.0 m/s 1.00 s later from a roof of height 95.0 m. (a) Where and when do they meet? (b) What are their velocities when they meet?

6) The magnitudes of the vectors **C** and **D** shown in the figure below are C = 4.0 m and D = 2.5 m. Find graphically: (a) **C** + **D**; (b) **C** – **D**.



7) Two vectors have equal magnitudes of 2.0 m. Find graphically the angle between them if the magnitude of their resultant is (a) 3.0 m; (b) 1.0 m. In each case use the law of cosines to confirm your answer.

8) A rectangular coordinate system with axes x' and y' is rotated by angle θ from axes x and y as shown in the figure. (a) What are the components of the position vector $\boldsymbol{r}$ in the two coordinate systems? (b) Use the results of part (a) to shown that the coordinates of a point $\boldsymbol{P}$ in the two systems are related by

$$x' = x\cos(\theta) + y\sin(\theta)$$
$$y' = -x\sin(\theta) + y\cos(\theta)$$

Hint, it might help to expand $\cos(\phi - \theta)$ *and* $\sin(\phi - \theta)$.